FasalSaakshi

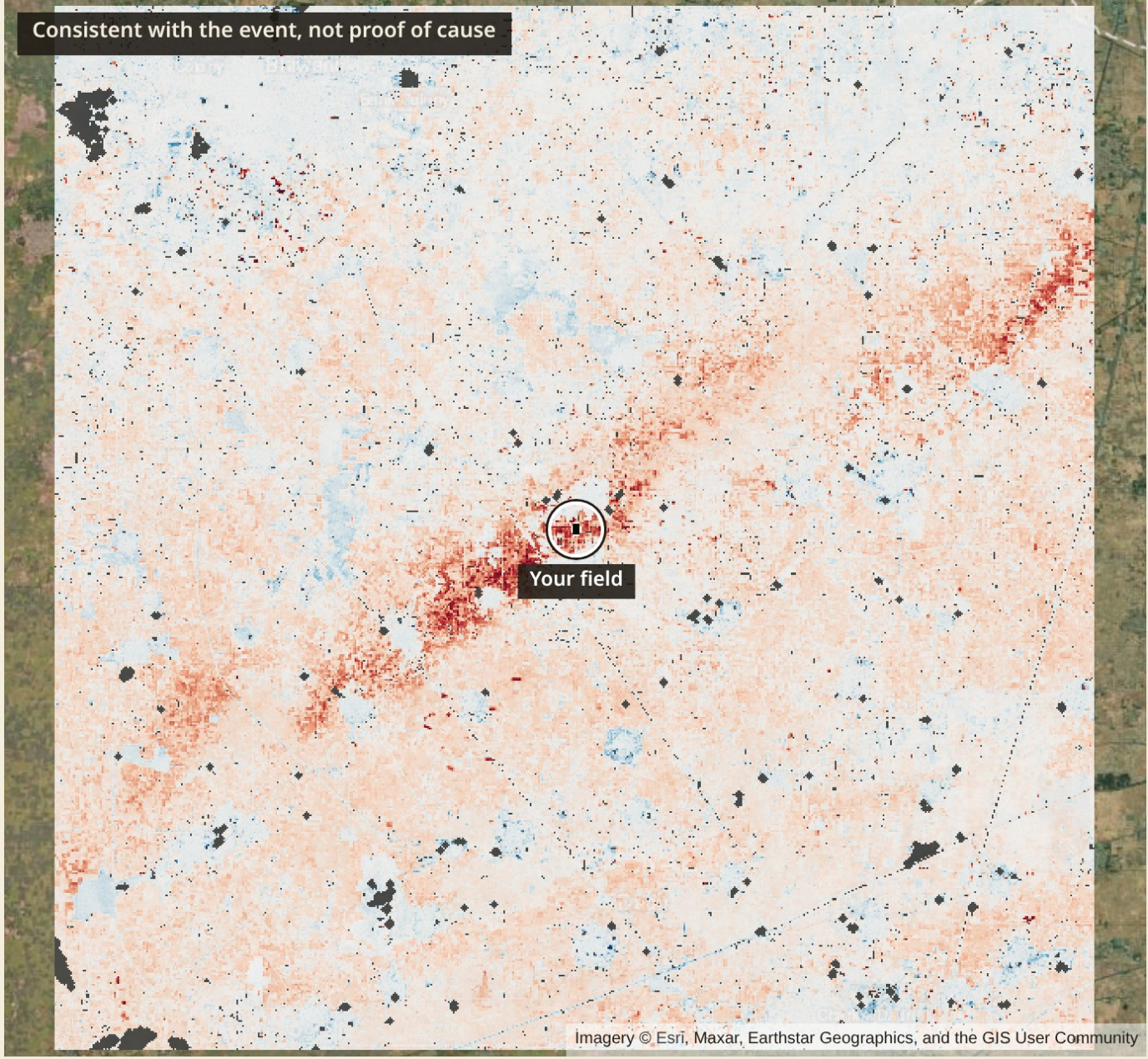
फसल साक्षी

A witness for the farmer's field.

A field's own satellite evidence, which anyone can recompute.

Team FasalSaakshi
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Forge '26 Gen AI Buildathon · Theme 2, Problem 2



Hail of 2-3 March 2024, Bhiwani. A band consistent with a hail swath, not proof of cause. **Marked plot: strongest of 200 scanned.**

Farmers can contest, but they have no evidence of their own to contest with

92.46 crore

"more than 92.46 crore farmer applications have been insured" since Kharif 2016. Applications are not farmers.

₹4,107.47 crore

of claims pending, as on 31.12.2025. Official causes are mainly State share, bank data and yield disputes.

About ₹350 crore

In 2025, Bhiwani and Charkhi Dadri farmers sat on dharna from 16 July. They say satellite-based yield cut about ₹350 crore from CCE-based claims.

Two systems, kept apart

YES-TECH, basic cover. Yield estimated "at Gram Panchayat (GP) level", rolled out from Kharif 2023. This is area yield, not a survey of one claim.

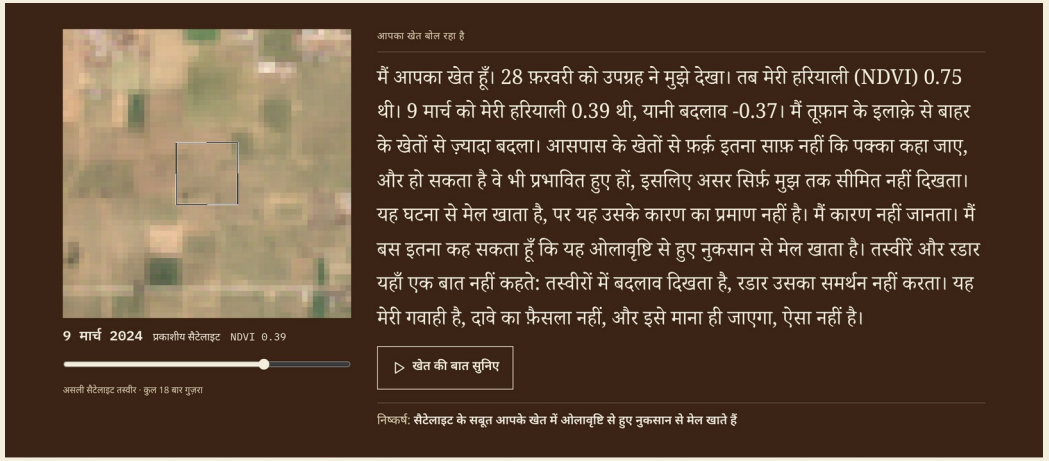
Localized claims, individual farm. A joint survey, or above 25% of insured area under intimation, a sample of 10 randomly selected parcels. Satellite indices enter only in a dispute.

Sources: PIB backgrounder, 29 Aug 2026; Lok Sabha Unstarred Q662, 3 Feb 2026, Annexure-I; Lok Sabha Q3825, 11 Aug 2026 (causes); The Tribune, 25 Aug 2025 (what farmers say, not confirmed); PMFBY Operational Guidelines 2023, paras 2.17, 21.5.8, 21.5.9.4. All via docs/submission/facts.md.

A witness who can say "your field doesn't show this"

The field testifies

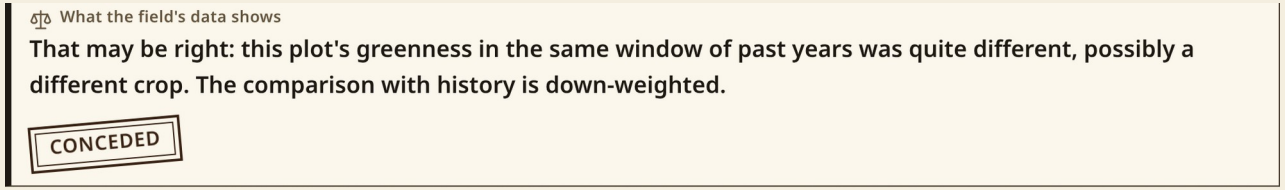
Its own record, with its caveats, in Hindi, Telugu or English.



The testimony, in Hindi. **Strongest of 200 scanned.**

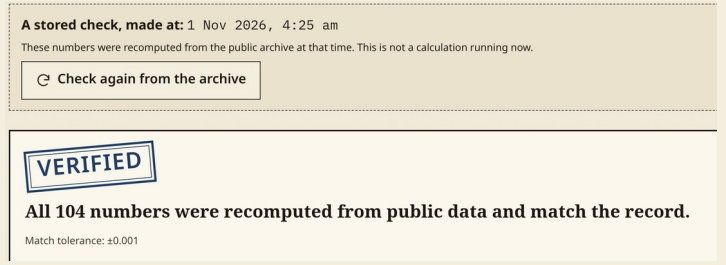
The surveyor cross-examines

Nine objections, tested on real values. It can concede: typical plot.



Anyone can recompute it

Public recompute of satellite numbers. Verdict, weather: tamper check.



A stored full check. **Strongest of 200 scanned.**
Separate clean-database runs gave identical hashes on all three records.

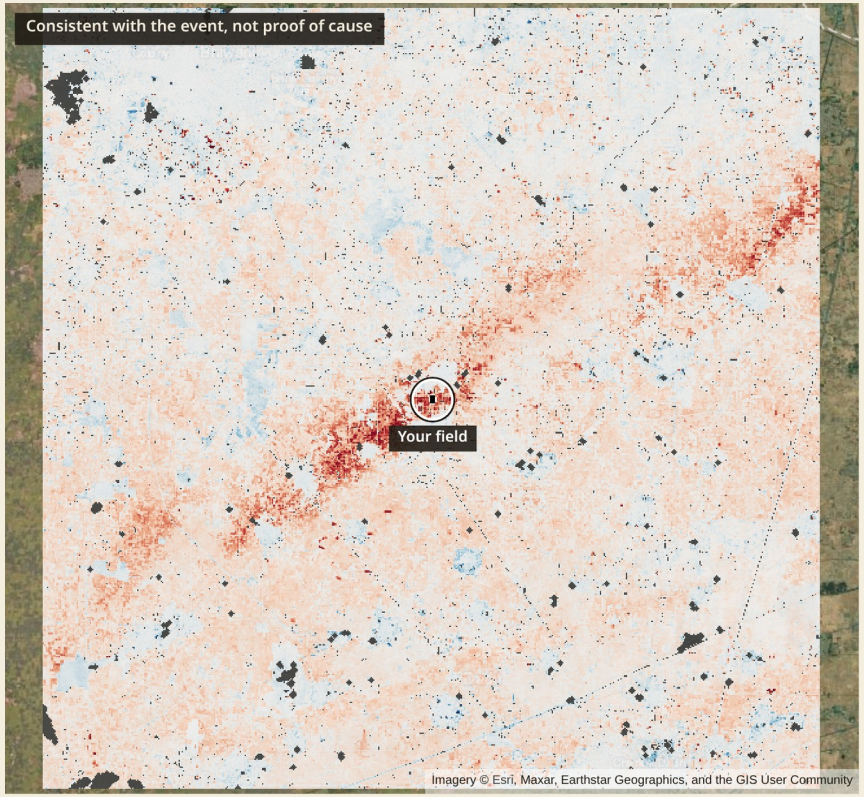
Gemini understands, explains and questions. Code measures and decides. A model can't talk the verdict up.

- Understands Hindi, Telugu and Hinglish reports
- "Ask the witness" answers from the farmer's own record, with a number check
- Flags contradictions as a question
- Advisory second opinion on images
- Reads the testimony aloud
- Finds related reports

Status: the live app runs on Gemini 3.6 Flash. The demo film's AI answers came from Claude standing in, and it says so on screen.

Sources: live sealed records /case/1ce7e64b754248abb18f0d29ac6203eb and /case/c157d2d75ce94501ad08930624a10e76; /verify/6092fe1c9ef12f2933ac9aa7fbc8b0737e9a306fdd6a071b159e64ea4dd7c58b. Identical hashes: README.md, final sealed records. Gen AI: docs/plan/07-build-contract.md Addenda 8 and 9.

Two questions, and what a wider average keeps



Hail of 2-3 March 2024, Bhiwani. Consistent with a hail swath, not proof of cause. **Plot: strongest of 200 scanned.**

Unlike comparable fields outside the storm?

Yes

0 of 200 changed as much. p = 0.005

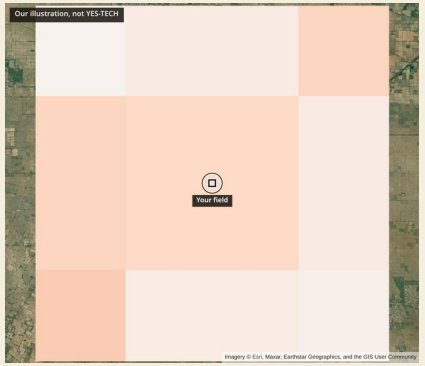
Unlike its own neighbours?

No, but near the line

11 of 200 changed as much. p = 0.0597

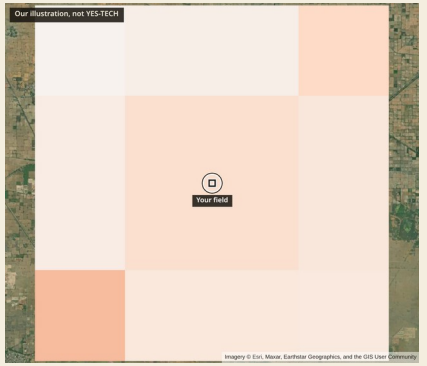
The Vanishing: same change, 3 km cells. Our illustration, not YES-TECH.

Strongest of 200 scanned



Field −0.369
3 km −0.098
27% remains
Verdict: strong

Typical, median of original 40



Field −0.126
3 km −0.081
64% remains
Verdict: weak

Ten-Field Draw, bounds not an estimate: −0.07 for all cropland, −0.245 for only fields that visibly dropped. The average depends heavily on who is on the list, and nobody outside the insurer can see that list.

Corroboration on the strongest plot: modelled rain of 18.4 mm in a day, with zero hail-code hours (live record).

Sources: sealed records 6092fe1c (strongest of 200 scanned, 1.27 ha; weather from its evidence record) and 56aadcb6 (typical, 1.09 ha), live app; Draw medians across 40 in-band plots, docs/validation/engine-results.md section 3e; Addendum 6. Imagery: Sentinel-2; base map credited on image.

One container. An explicit agent. Code decides.

Farmer's phone → React + MapLibre web app → FastAPI (one URL, live trace over SSE) → LangGraph state graph

The graph, with real branches

parse → plot → weather and alerts → optical search and QA → compare, or switch to radar → history → cross-examine (can loop back) → verdict → route → explain → exhibits → seal

Measures and decides: code

Sentinel-2 L2A and Sentinel-1 RTC, windowed reads from Planetary Computer, Earth Search as fallback.
Scene QA. Placebo tests on same-shape plots, near and 5-12 km away.
Fixed verdict rubric. Weather from Open-Meteo and NASA POWER, alerts from NDMA SACHET.

Understands and explains: LLM

Report parsing, Ask the witness, contradiction question, advisory image reading, speech.
Number and phrase checks, grounded in the case record and clauses from PMFBY Guidelines 2023.
Stored outside the seal. Falls back to fixed text. It can't change a verdict.

Seals and proves

SHA-256 over polygon, scene ids, seed, engine version, numbers, and a digest of verdict and weather.
A public page recomputes satellite numbers from the archives, caches bypassed.
Separate clean runs gave identical hashes.

Python, FastAPI, LangGraph, rasterio, numpy, shapely, SQLite or Postgres, WeasyPrint, one Docker image. Source and setup: github.com/AnirudhSinghRajora/Forge

What we measured, and what we didn't

False alarms, two locked no-event windows	**2.0-2.5%**	2.0% strong, 2.5% moderate field conditions, 200 plots per window, rubric frozen beforehand. Each window is one scene pair, so the CIs are too narrow.
Independent event: Sirsa hail, 14 Mar 2025	**0 of 200**	Not a pass. The Tribune called the damage "minor", and we cannot tell little damage from a miss.
Punjab floods 2025, pre-registered	**Partial**	New water on 28.3% of plots against 0.0% in the control: passes. Radar placebo 1.7% against 8.3%: fails.
Detection through the field test	**Bhiwani only**	Shown only on Bhiwani, the event the far test and post-scene rule were designed on. Radar there does not confirm it: the orbits disagree.
Verdicts near the threshold	**Fragile**	Near p 0.025 the level flipped on up to a third of 30 seeds, on plots picked for being near that line. A slightly different drawing can change the result. Clear cases held.
Ground truth	**None**	No plot-level ground truth for any event. This work shows consistency, not accuracy.

Sources: docs/validation/engine-results.md sections 3c, 7, 7a, 7b, 7c; docs/plan/07-build-contract.md Addendum 7. CIs: 0.8-5.0% strong, 1.1-5.7% moderate. Field conditions: the plot rule without corroboration. Thirty seeds do not bound another seed: the sealed strongest plot's ring p 0.0597 sits just above that table's range, with no change of level (7c).

Who this could help, and what comes next

241.38 lakh

farmers insured in Kharif 2026, as of 27 Aug 2026.

Of 2024-25 farmer applications, 17.6% are marginal, 6.5% tenant and 48% loanee farmers.

74% of the ₹4,107.47 crore pending is in Andhra Pradesh, where Arnab just hit. That is not a claim that satellites caused it, and this does not fix a State's unpaid share.

No measured farmer outcomes yet. No insurer or State has reviewed this.

Anyone can check what it said.

Next, in order

1. Re-check Arnab at the next clear passes: Sentinel-2 about 2 Nov, radar about 4 Nov. No signal is established yet.
2. Test sensitivity on an independent event with real damage.
3. Test against real records: crop-cutting or joint survey results for specific plots.
4. Complete the Gemini run, and get Telugu reviewed by a native speaker.
5. Pilot with an FPO or KVK. Ask an insurer and a State team what they'd need.
6. Verify State notifications, so routes stop saying "confirm with 14447".

Try it live

exercise-instance-promptly-proxy.trycloudflare.com

Sources: PIB backgrounder, 29 Aug 2026; PIB 2148513, 25 Jul 2025; Lok Sabha Unstarred Q662, 3 Feb 2026, as on 31.12.2025, our arithmetic (₹3,041.66 crore), all via docs/submission/facts.md sections 2, 3, 5. Radar date: docs/validation/engine-results.md section 4.